Madhav Chopra

Senior Associate | Full-Stack Developer & Python Specialist

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Summary

Experienced Python Developer with more than 5 years of expertise in full-stack automation, system migrations, and data engineering. Skilled in leading cross-functional teams to streamline data extraction, sanitization, and dynamic web application development using modern Python frameworks (BeautifulSoup, Polars).

Skills

Languages: Python (3.12+), JavaScript, SQL, Bash/Shell Scripting, HTML5, CSS3

Frameworks & Web: FastAPI, Django, ReactJS, Next.js, Pydantic, Selenium, Playwright, Beautiful Soup

Data & ML: Polars, Pandas, NumPy, Scikit-Learn, NLTK, Matplotlib, Seaborn, Jupyter

Cloud & DevOps: Docker, Docker Compose, Ansible (IaC), Azure DevOps, Git, Nginx, Linux Administration

Virtualization: Proxmox VE, LXC/LXD Containers, GPU Passthrough, Portainer

Core Engineering: uv (Package Management), Multithreading, Asyncio, REST APIs, System Design, Agile

Experience

Senior Associate, PwC India – Gurugram, Haryana Apr 2026 – present

Backend engineer specializing in the FastAPI ecosystem, delivering end-to-end backend solutions from initial design through deployment for projects including APMT and SEPS.

- Architected and delivered production-grade REST APIs using **FastAPI** for Adani's **APMT** and **SEPS** projects, providing complete backend support.
- Designed scalable backend architectures leveraging **Pydantic** for data validation, **asyncio** for high-throughput request handling.
- Collaborated closely with cross-functional teams to translate business requirements into technical solutions, maintaining high code quality and adherence to **Agile** best practices.

System Engineer, Tata Consultancy Services – Gurugram, Haryana July 2021 – Apr 2026

Hand-picked for the Rollout Team to lead high-stakes website migration initiatives for the Adecco client, architecting Python-based ETL pipelines for automated data extraction and dynamic sanitization with 100% data integrity. Also delivered full-stack feature enhancements for enterprise-scale web applications, serving as a key liaison between technical teams and international clients.

- Engineered a migration automation suite using **Polars**, **Pandas**, and **BeautifulSoup** to automate data extraction, transformation, and dynamic sanitization, slashing delivery timelines by **50%** while doubling data processing throughput.
- Developed a custom **SEO Checker** script to dynamically validate migration accuracy, generating automated Excel reports that reduced manual QA time by **40%**.
- Collaborated with cross-functional teams to build end-to-end critical features utilizing **ReactJS** and **Python**, delivering a modular, responsive, and scalable codebase that improved user engagement across devices.
- Led performance optimization to boost **Core Web Vitals** and achieve **WCAG accessibility compliance** recognized as a **"Presentation Hero"** for exceptional stakeholder management, simplifying complex technical updates for international clients.

Projects

SEO & Link Compliance Auditor

- Architected a cross-platform desktop application for batch SEO auditing, utilizing **Playwright** for robust asynchronous scraping and dynamic content rendering.
- Optimized runtime performance by implementing an intelligent **link caching system** and URL normalization logic to eliminate redundant network requests.
- Developed a dual-interface system (CLI for CI/CD, Tkinter GUI for users) supported by comprehensive **Pytest** coverage.
- Automated the generation of professional-grade Excel reports using **Pandas** and **OpenPyxl** with conditional formatting for instant analysis.

Automated Infrastructure

Designed and maintained a enterprise-grade lab environment for testing and deployment.

- Deployed **Ansible** playbooks to automate configuration management across Linux LXC containers.
- Managed containerized services using **Docker Compose** and **Portainer**, ensuring high availability for client services.
- Hardened server security using **UFW** firewalls, SSH key-based authentication, and reverse proxy management (Nginx).

Education

Punjab Technical University, Bachelor of Technology in Computer Science and Engineering — May 2021

Certifications

Generative AI & LLMs: Azure OpenAI & GenAI Models, Prompt Engineering, Building Custom GPTs

Data Science & ML: OpenEDG Python Professional, Scikit-Learn, Pandas to Polars, Matlab ML

Cloud & DevOps: Azure DevOps Fundamentals (Udemy), DevOps Foundations, Git Essential Training

Programming & Tools: Linux Shell Scripting, JavaScript (ES6+), Vim, Machine Learning (Stanford/Coursera)

Awards

Beyond Performance Award — Sep 2024 & Jan 2023

Service & Commitment Award — July 2024

Star of the Month — Apr 2022